Alberto Hernández Espinosa
Mexico City, Mexico • +52 552 383 4438 • albertohernandezespinosa@gmail.com
https://albertoHdzE.github.io/website/ • https://linkedin.com/in/alberto-hernandeze • https://github.com/albertoHdzE

1 Professional Summary

- AI researcher with over 10 years of experience leading scientific software development for universities and industry, specializing in artificial intelligence, cognitive science, and complexity science.
- Expertise in designing and deploying machine learning, deep learning, and causal AI models, with a proven track record of delivering impactful solutions (e.g., 53% cloud cost reduction).
- Published with Oxford University and Oxford Immune Algorithmics, contributing to advancements in artificial intelligence, integrated information theory and algorithmic complexity.
- Skilled in remote collaboration, having worked globally for 7 years, and prepared to work with interdisciplinary AI teams.
- AI Engineer Lead at K-Square Group, driving AI integration and fostering innovation through agentic platforms and research.

2 Skills & Expertise

- AI/ML Technologies: Python, TensorFlow, PyTorch, Scikit-learn, Keras, LangGraph, Ollama, ML Studio, RAG, Agents, MCP, Small Model Tuning, Multi-Agent Orchestration, Local LLM Orchestration, Tool-Use LLMs, Chain-of-Thought Prompting, Self-Reflection LLMs, LLM-as-a-Judge Patterns, Langfuse
- Programming Languages: Python, R, C++, SQL, Mathematica, FastAPI, React
- Cloud & DevOps: GCP, AWS, Docker, FastAPI, Telemetry, Monitoring, Observability
- Data Science: Big Data, SQL/NoSQL, ETL, Pandas, NumPy, Matplotlib
- Domains: Artificial Intelligence, Complexity Science, Cognitive Science, Reinforcement Learning, Algorithmic Information Theory, Enhancing RAG with Complexity, Theoretical Limits on Intelligence of LLMs, Agentic LLM Reasoning, Emergent Collective Intelligence, Quantitative Trading, Agentic Trading Systems, Deterministic Strategy Generation

3 Professional Experience

3.1 AI Engineer Lead, K-Square Group (Remote)

Nov 2024–Present

- Conduct and lead teams for implementation of AI-driven projects, enhancing team collaboration and project execution.
- Spearhead research and development of AI/ML applications tailored to specific industry challenges, focusing on innovation and efficiency.
- Design and oversee the integration of advanced AI technologies, including agentic systems and complex data processing pipelines.
- Develop and implement strategic frameworks for rapid prototyping and proof-of-concept development, aligning with organizational goals.

- Foster a culture of experimentation and continuous improvement, mentoring teams in leveraging AI tools effectively.

3.2 Chief AI Scientist, MilkStraw AI, San Francisco, CA (Remote)

Oct 2022–Sep 2024

- Led research of AI-driven cloud resource optimization models on AWS, achieving up to 53% cost savings for clients.
- Designed and deployed ML pipelines using Python, TensorFlow, and PyTorch.
- Collaborated remotely with global teams to deliver scalable AI solutions.

3.3 Research Director, Laxford Capital (Remote)

Jun 2019–Jul 2022

- Directed AI trading initiatives, developing reinforcement learning agents for financial markets.
- Oversaw model design and implementation in Python and PyTorch.
- Enhanced trading performance through causal AI approaches.

3.4 Research Collaborator, Oxford Immune Algorithmics & Oxford University, Oxford, UK (Remote with Visits)

2015–Nov 2024

- Contributed to publications on integrated information theory and algorithmic complexity.
- Implemented models in Mathematica and Python.
- Co-authored papers with Oxford researchers, advancing cognitive science applications.

4 Education

4.1 PhD in Philosophy of Science, UNAM, Mexico City, Mexico

2015–2019 (Specialization: Artificial Intelligence and Cognitive science)

- Research focused on integrated information theory (Phi).
- CONACyT Scholarship.
- Education in association with University of Oxford, Computer Science Faculty.

4.2 Master in Computer Sciences, UNAM, Mexico City, Mexico

2012–2014 (Specialization: Artificial Intelligence)

- Thesis on computational creativity models.
- CONACyT Scholarship.

5 Selected Projects

- GraphRAG From Scratch
  - Clean, mathematically explicit implementation of RAG, RAU, and GraphRAG for scientific comparison.
  - Exposes embeddings, graph construction, traversal, and synthesis as separate components to study when graph-based retrieval improves multi-hop reasoning.
  - Code: https://github.com/albertoHdzE/GraphRAG-scratch
- HiFi
  - Multi-agent automatic trading system built on a local agentic architecture that tunes different families of local LLMs through ML Studio for deterministic strategy creation.
  - Integrates MCP, tools, ML/AI pipelines, ContextForge, Langfuse, deep telemetry, and monitoring for observable orchestration, evaluation, and control of trading workflows.
  - Code: https://github.com/albertoHdzE/HiFi
- CouncilAI V2 (AI POCs Feature Spanner)
  - Reproducible benchmarking harness and dashboard for comparing vector, graph, long-context, and managed-search retrieval substrates under a fixed agentic workflow.
  - Measures latency, verifiability, and retrieval quality across multiple RAG pipelines using deterministic run artifacts and a React dashboard.
  - Code: https://github.com/albertoHdzE/ai-pocs-feature-spanner
- ContextForge (k-mcp-farm)
  - Open-source registry and proxy that federates MCP, A2A, and REST/gRPC APIs with centralized governance, discovery, and observability.
  - Consolidates tools, agents, and APIs behind a clean endpoint for AI clients, with plugin extensibility and Kubernetes-ready deployment.
  - Code: https://github.com/albertoHdzE/k-mcp-farm
- AG-UI
  - Pluggable agent-user interaction layer for human-machine communication in agentic applications.
  - Supports interactive interfaces that make complex AI workflows more observable and usable.
  - Code: https://github.com/albertoHdzE/agui
- CausalBool: Deterministic Causal Integration
  - Physics-inspired algorithmic information theory for Boolean networks, deriving exact closed-form formulae for causal reconstruction without probabilistic assumptions.
  - Implemented deterministic index algebra and ordering-aware invariances to scale information signatures (PID, Transfer Entropy) beyond combinatorial limits.
  - Code: https://github.com/albertoHdzE/CausalBool
- Noesis: Agentic Agnostic Platform
  - Platform for creating AI/ML models (regression, classification, clustering, anomaly detection) using LangGraph, Ollama, local tuned models, FastAPI, and React.
  - Applied to Fraud Detection in financial and insurance institutions.
  - Code: https://github.com/KSProjectX/test-xx (restricted access).

- KS-Onboarding: Agentic Platform for Project Control and Onboarding
  - Deep agents for organizing projects, analyzing documentation/meetings, generating insights, and responding to questions using local/external information, OCR, and voice recognition.
  - Advanced RAG enhanced with complexity science based on high-level research (https://arxiv.org/abs/2505.02581).
  - Code: https://github.com/KSProjectX/KS-onboarding (restricted access).
- Agents for Automation of Running Tests
  - Agentic platform using user stories/use cases in natural language or code to dynamically create pytest test code and report coverage.
  - Code: https://github.com/Sucharita8/KS-qe-platform (restricted access).
- Stock Prices Prediction
  - Developed predictive and autonomous decision-making models for stock prices based on Deep Reinforcement Learning and advanced ML approaches.
  - Integrated time-series analysis and reinforcement learning.
  - Deployed on cloud platforms for real-time forecasting.
  - GitHub: https://github.com/albertoHdzE/finance_1, Detailed explanation: https://sites.google.com/view/complexai/FINANCE.
- MOSAICA: DNA Complexity Analysis
  - Investigated DNA sequence complexity using Machine learning with Python and Mathematica.
  - Correlated algorithmic complexity with DNA curvature.
  - Results published in academic papers.
  - GitHub: https://github.com/albertoHdzE/MOSAICA.

6 Publications

- 2025: "Neurodivergent Influenceability as a Contingent Solution to the AI Alignment Problem." PNAS. https://academic.oup.com/pnasnexus/article/5/4/pgag076/8651394
- 2025: "SuperARC: An Agnostic Test for Narrow, General, and Super Intelligence Based On the Principles of Recursive Compression and Algorithmic Probability." Nature. https://www.nature.com/articles/s41467-026-73289-5
- 2021: "Estimations of Integrated Information Based on Algorithmic Complexity and Dynamic Querying." World Scientific. https://www.worldscientific.com/doi/10.1142/9789811235726_0005
- 2017: "Is There Any Real Substance to the Claims for a 'New Computationalism'?" Springer. https://link.springer.com/chapter/10.1007/978-3-319-58741-7_2
- 2013: "Does the Principle of Computational Equivalence Overcome the Objections Against Computationalism?" Springer. https://link.springer.com/chapter/10.1007/978-3-642-37225-4_14

7 Certifications

- Mathematics for Machine Learning
- AI for Finance
- Google Cloud Specialization
- Reinforcement Learning
- Complexity Science

Full list of 47 certifications available here: https://www.linkedin.com/in/alberto-hernandeze/details/certifications/

8 Languages

- Spanish (mother language)
- English (advanced proficiency)
- German (Basic)
- French (Basic)
- Japanese (Basic)

Certifications available here: https://www.linkedin.com/in/alberto-hernandeze/details/languages/